

Processing PACE OCI data using NASA's SeaDAS

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Overview

- Data access options on the PACE website
- Installing SeaDAS + Docker
- Getting L2 PACE OCI data from NASA EarthData
- Processing that L2 data to L3 using SeaDAS

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Access PACE Data

Public PACE data release began on 11 April 2024. **Reprocessing Version 3 is now available.**

Reprocessing Version 3 is the second full mission reprocessing, and primarily serves to incorporate improved refinement of the calibration for the three PACE instruments, as well as various algorithm refinements, bug fixes, data format improvements, and expanded product suites. The reprocessing includes all standard science mode data collected during the PACE commissioning period, starting from instrument engineering first light in February 2024. As for previous reprocessings, we welcome your input and discoveries, but also request your continued patience while we continue to improve the data and implement advanced algorithms. Frequent [updates](#) and [reprocessings](#) to incorporate post-launch calibration knowledge, algorithm refinements, and additional data products should be expected.

PACE Data Resources

- [Release notes for Version 3](#)
- [Release notes for Version 2](#)
- [Release notes for Version 1](#)
- [A complete list of science data products](#), including maturity levels and the status of current and pending data availability for each product
- [Information on working with PACE data](#)
- [Learn what you should know about PACE data](#)

Options for accessing PACE data

PACE data are accessible through several options described on the [Ocean Biology \(OB\) DAAC Find Data](#) and [NASA Earthdata](#) web sites.

Three primary options include:

- [Earthdata Search OB.DAAC portal](#)
- [OB.DAAC Level 3 & 4 Browser](#) (Note: Within the "Product Status" pulldown select "Provisional" or "Testing" to view data.)
- [OB.DAAC File Search](#)

The OB.DAAC Level 1 & 2 browser does not support access to PACE data.

[PACE orbit lines](#), [true color](#), and [chlorophyll data](#) are also viewable on [Worldview](#).

Access by Maturity Level

Access to data varies with [data maturity level](#). Level-1 data from OCI, HARP2, and SPEXone are classified as Provisional. The limited suite of OCI Level-2 and -3 derived products are classified as Provisional, Diagnostic, and/or Test.

- Provisional Level-1 and -2 data are available through Earthdata Search and the OB.DAAC File Search. Provisional Level-3 data are available via all three options. Note that only Provisional data are available through Earthdata Search.
- Test and Diagnostic Level-2 data are available through the OB.DAAC File Search. Test and Diagnostic Level-3 data are available through the OB.DAAC Level 3 & 4 Browser and File Search. Note that Test and Diagnostic data are not available through Earthdata Search.

Summary of currently available products

V3 OCI products include:

- Level-1A/B/C radiometric data (Provisional)
- The ocean apparent optical properties suite (OC_AOP): spectral remote sensing reflectances and uncertainties (Provisional), apparent visible wavelength (Provisional), normalized fluorescence line height (Provisional), aerosol optical thickness (Diagnostic), and aerosol angstrom exponent (Diagnostic). Note that these aerosol properties represent the atmospheric correction

https://pace.oceansciences.org/access_pace_data.htm

https://seadas.gsfc.nasa.gov/client_server/

EARTHDATA

Other DAACs ▾



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SEADAS

SeaDAS-OCSSW Client Server Configuration

Introduction

The SeaDAS processing components (OCSSW) can be installed only on Linux or macOS (Intel) systems. For this reason, it has been difficult for Windows users to use the OCSSW programs in an efficient way. SeaDAS 8.1.0 and future releases use a client/server model, which enables the physical decoupling of the SeaDAS GUI application and the OCSSW package. (Note SeaDAS 8.0.0 was a Beta version and did not support client/server model. SeaDAS 7.5 also supports client/server model)

Since the SeaDAS client and the OCSSW server reside on two separate machines, they need a way to share files. They can either transmit the files over the network or use a shared folder. For virtual machines, we can use a shared folder and eliminate the transmission of files over the network.

https://seadas.gsfc.nasa.gov/client_server/

Click for instructions:

Docker

Virtual Machine

Windows Subsystem for Linux

Remote Machine

1. Follow these instructions to install [Docker Desktop](#).
2. Please make sure that you have created a valid [.netrc](#) file in your home directory (\$HOME) and also copy it into the \$HOME/seadasClientServerShared/ directory..

3. Under your

\$HOME

directory, create the

"ocssw"

and

"seadasClientServerShared"

directories if they don't already exist. (NOTE: You can create these directories anywhere that you have full permission to operate. You have to provide the full path later)

For example,

```
> cd
```

```
> mkdir ocssw
```

```
> mkdir seadasClientServerShared
```

4. Make sure Docker Desktop is running before running the following command in Windows Power Shell's command line:

```
docker run -v $HOME/ocssw:/root/ocssw -v $HOME/seadasClientServerShared:/root/seadasClientServerShared -p 6400:6400 -p 6402:6402 -p 6403:6403 -t seadas/ocssw-run:2.0
```

Skip to Section 4: [SeaDAS Client Configuration](#).

Section 4: SeaDAS Client Configuration

To configure the client to communicate with the server, the SeaDAS config file needs to be edited.

```
${user.dir}\.seadas9\etc\seadas.properties
```

Click an environment to see which lines to edit in the *seadas.properties* file:

Docker	Virtual Machine	Windows Subsystem for Linux	Remote Machine
<pre>seadas.ocssw.location=docker seadas.ocssw.port=6400 seadas.ocssw.sharedDir=C:\\Users\\\${user.name}\\seadasClientServerShared seadas.client.id=\${user.name} seadas.ocssw.keepFilesOnServer=false seadas.ocssw.processInputStreamPort=6402 seadas.ocssw.processErrorStreamPort=6403</pre>			

 Ask Gordon **BETA**

 Containers

 Images

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 MCP Toolkit **BETA**

 Docker Hub

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 Extensions

Containers [Give feedback](#)

View all your running containers and applications. [Learn more](#)

Container CPU usage 

0.09% / 2000% (20 CPUs available)

Container memory usage 

317.2MB / 30.46GB

[Show charts](#)

 Search



Only show running containers

Delete



☒	Name	Container ID	Image	Port(s)	Actions
☒	 optimistic_kirch	ae2827915f41	seadas/oc	6400:6400 Show all ports (3)	  

Selected 1 of 1

Walkthroughs



Multi-container applications

\$ docker init

Containerize your application

Search

...

[View: List](#) [Home](#) - [FOIA](#) - [NASA Privacy Policy](#) - [USA.gov](#)

[https://search.earthdata.nasa.gov/search?fi=OCI&as\[instrument\]\[0\]=OCI](https://search.earthdata.nasa.gov/search?fi=OCI&as[instrument][0]=OCI)

🔍 surface reflectance

Search

6 Matching Collections

Showing 6 of 6 matching collections

Sort: Usage

View: List

📅 Temporal ▾

📍 Spatial ▾

☰

🗑️

INST: OCI ✕

🔼 Filter Collections

Features

- ☐ ☁ Available in Earthdata Cloud
- ☐ ⚙ Customizable ⓘ
- ☐ 🗺 Map Imagery

Keywords ▾

Platforms ▾

Instruments 1 Selected ▾

Organizations ▾

Projects

Processing Levels ▾

Data Format ▾

Tiling System

Horizontal Data Resolution

Latency ▾

Additional Filters

- ☐ Include collections without granules
- ☐ Include only EOSDIS collections

PACE OCI Level-2 Regional Surface Reflectance Data, version 3.0

56,872 Granules 2024-03-05 to Present



No image available

The primary sensor aboard the PACE spacecraft is the Ocean Color Instrument (OCI). It is a highly advanced optical spectrometer that will be used to measu...

[GEOSS · PACE_OCI_L2_SFREFL v3.0 - NASA/GSFC/SED/ESD/GCDC/OB.DAAC](#)

📘 +

PACE OCI Level-2 Regional Surface Reflectance - Near Real-time (NRT) Data, version 3.0

27,288 Granules 2024-03-05 to Present



No image available

🕒 1 to 4 days

The Ocean Biology DAAC produces near real-time (quicklook) products using the best-available combination of ancillary data from meteorological and...

[GEOSS · PACE_OCI_L2_SFREFL_NRT v3.0 - NASA/GSFC/SED/ESD/GCDC/OB.DAAC](#)

📘 +

PACE OCI Level-3 Global Mapped Surface Reflectance Data, version 3.0

987 Granules 2024-03-05 to Present



No image available

The primary sensor aboard the PACE spacecraft is the Ocean Color Instrument (OCI). It is a highly advanced optical spectrometer that will be used to measu...

[GEOSS · PACE_OCI_L3M_SFREFL v3.0 - NASA/GSFC/SED/ESD/GCDC/OB.DAAC](#)

📘 +

PACE OCI Level-3 Global Mapped Surface Reflectance - Near Real-time (NRT) Data, version 3.0

504 Granules 2024-03-05 to Present



No image available

🕒 1 to 4 days

The Ocean Biology DAAC produces near real-time (quicklook) products using the best-available combination of ancillary data from meteorological and...

[GEOSS · PACE_OCI_L3M_SFREFL_NRT v3.0 - NASA/GSFC/SED/ESD/GCDC/OB.DAAC](#)

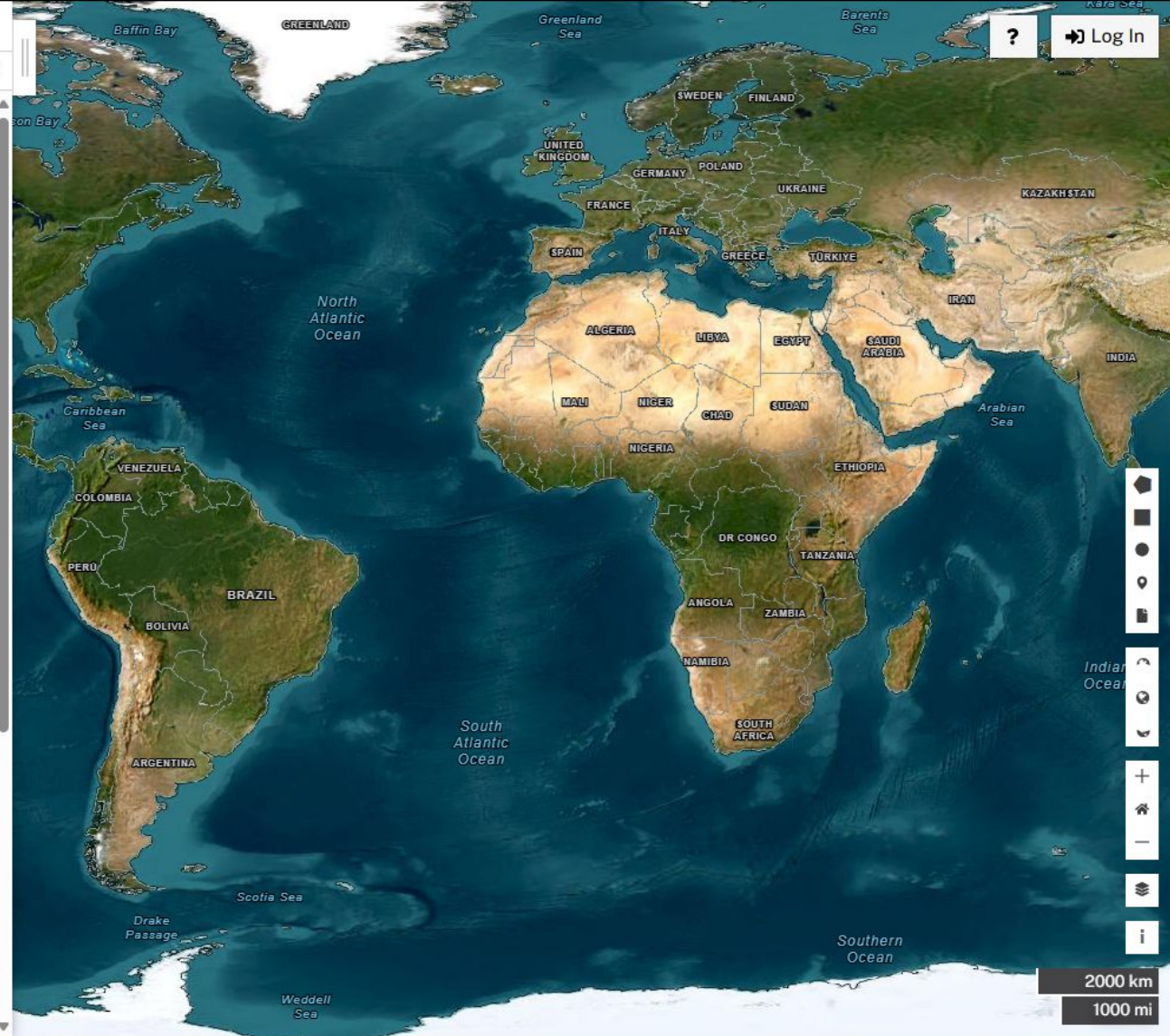
📘 +

PACE OCI Level-3 Global Binned Surface Reflectance Data, version 3.0

477 Granules 2024-03-05 to Present



No image available



🔍 surface reflectance Search

Temporal Spatial

INST: OCI

Spatial Rectangle

SW:

NE:

Filter Granules Clear Filters

Granule Search

Granule ID(s) @

Temporal

Start

YYYY-MM-DD HH:mm:ss

End

YYYY-MM-DD HH:mm:ss

☐ Use a recurring date range

Day/Night

Find granules captured during the day, night or anytime.

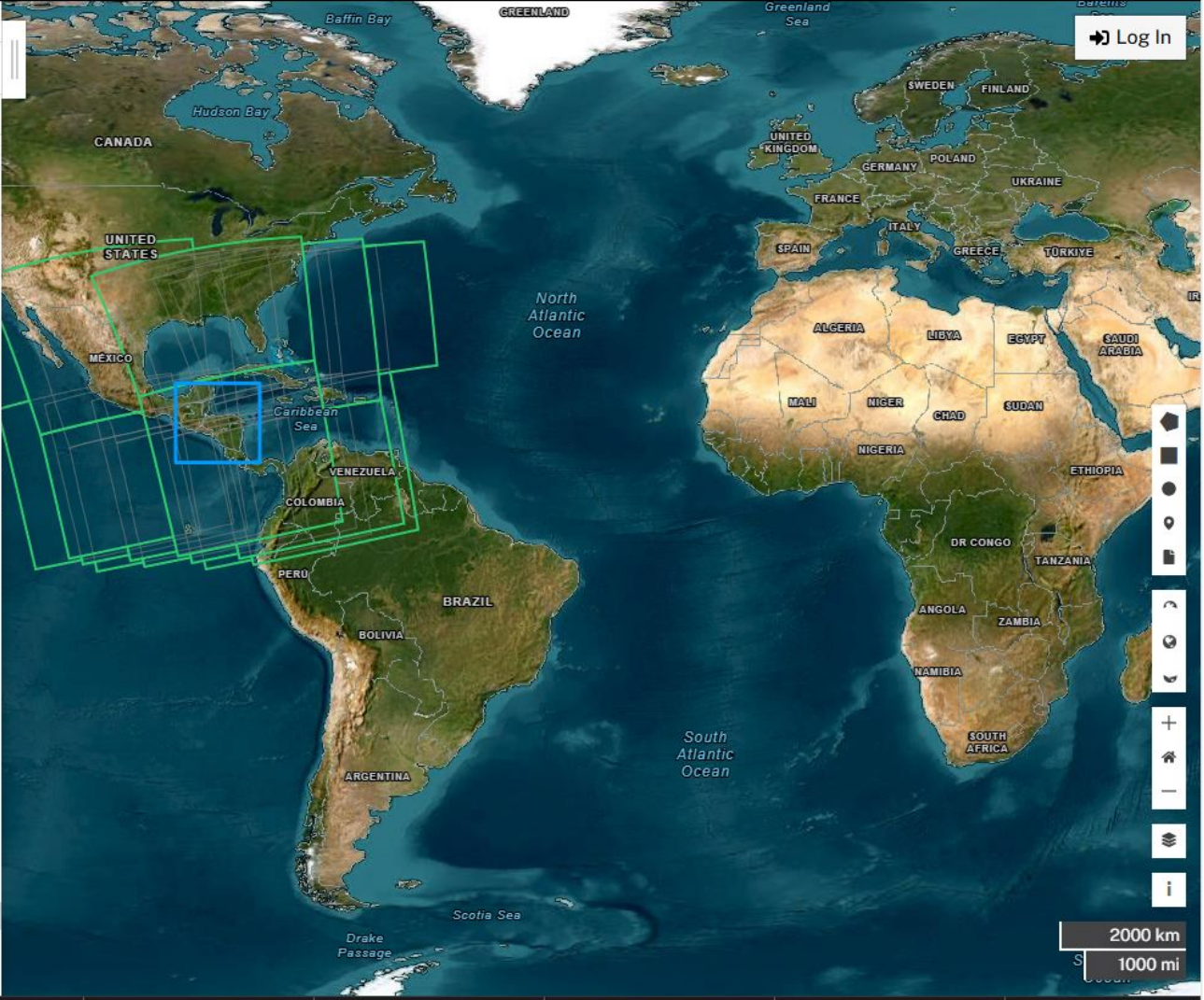
Search Results (6 Collections)

PACE OCI Level-2 Regional Surface Reflectance Data, version 3.0

Showing 20 of 985 matching granules Sort: Start Date (Newest) View: List

PACE_OCI.20250501T183011.L2.S FREFL.V3_0.nc START 2025-05-01 18:30:11 END 2025-05-01 18:35:10	PACE_OCI.20250501T182511.L2.S FREFL.V3_0.nc START 2025-05-01 18:25:11 END 2025-05-01 18:30:10
PACE_OCI.20250430T193312.L2. SFREFL.V3_0.nc START 2025-04-30 19:33:12 END 2025-04-30 19:38:11	PACE_OCI.20250430T192812.L2. SFREFL.V3_0.nc START 2025-04-30 19:28:12 END 2025-04-30 19:33:11
PACE_OCI.20250430T175451.L2.S FREFL.V3_0.nc START 2025-04-30 17:54:51 END 2025-04-30 17:59:50	PACE_OCI.20250430T174951.L2.S FREFL.V3_0.nc START 2025-04-30 17:49:51 END 2025-04-30 17:54:50
PACE_OCI.20250429T185252.L2. SFREFL.V3_0.nc START 2025-04-29 18:52:52 END 2025-04-29 18:57:51	PACE_OCI.20250429T171930.L2.S FREFL.V3_0.nc START 2025-04-29 17:19:30 END 2025-04-29 17:24:29

Add Download All 985



Earthdata Access: A Section 508 accessible alternative

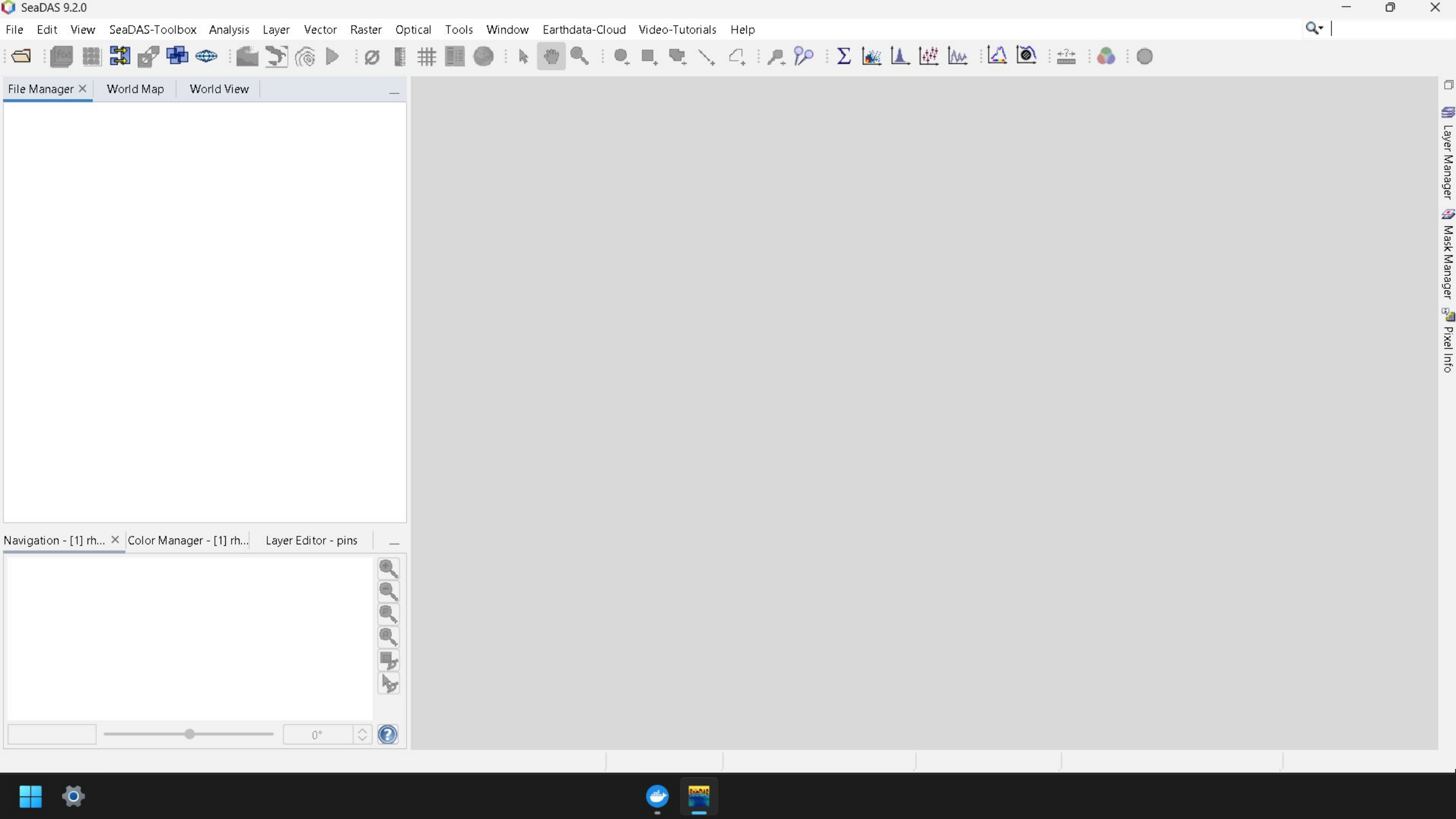
SeaDAS

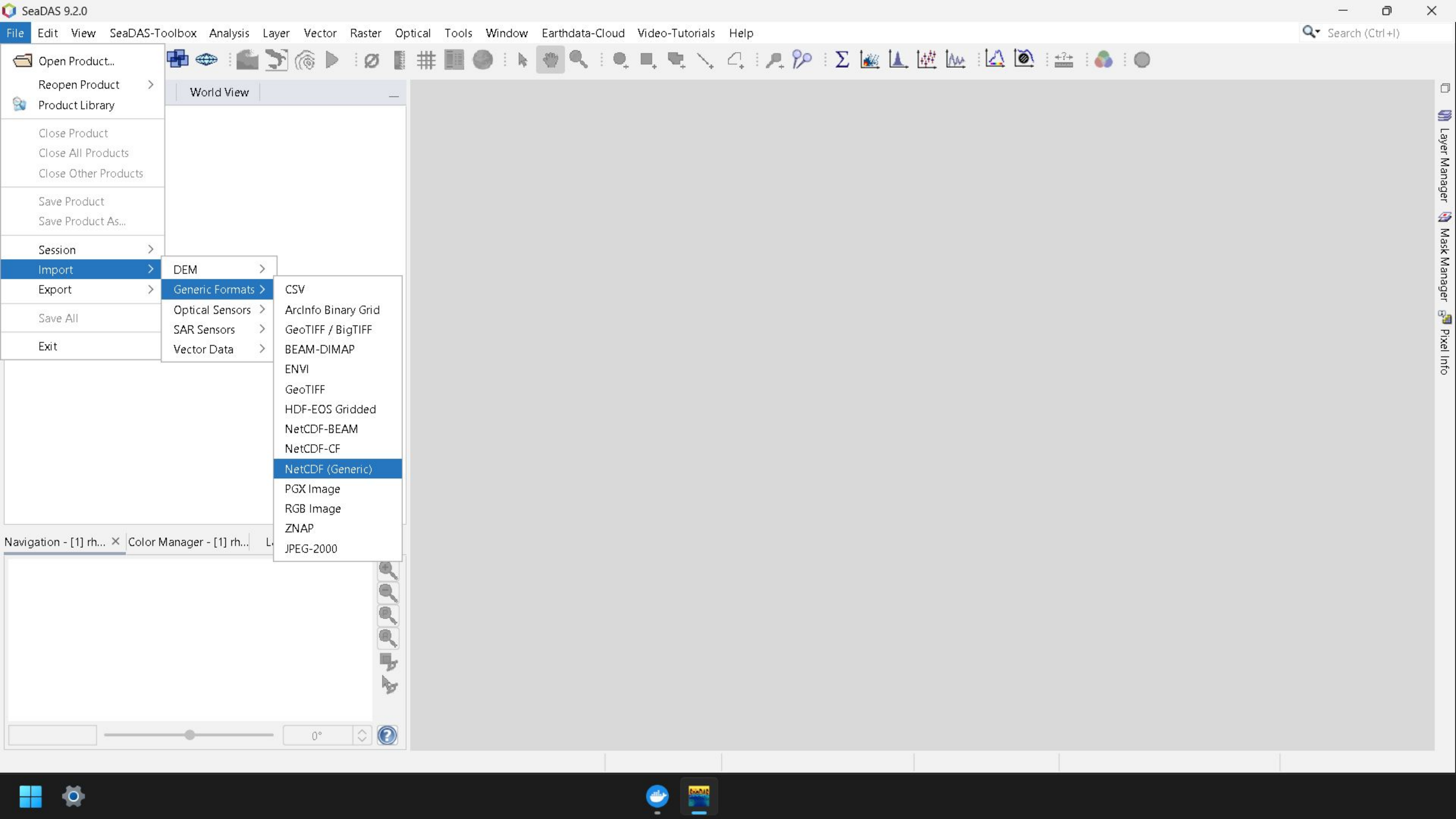
Done loading modules.



Sea, earth & atmosphere Data Analysis System

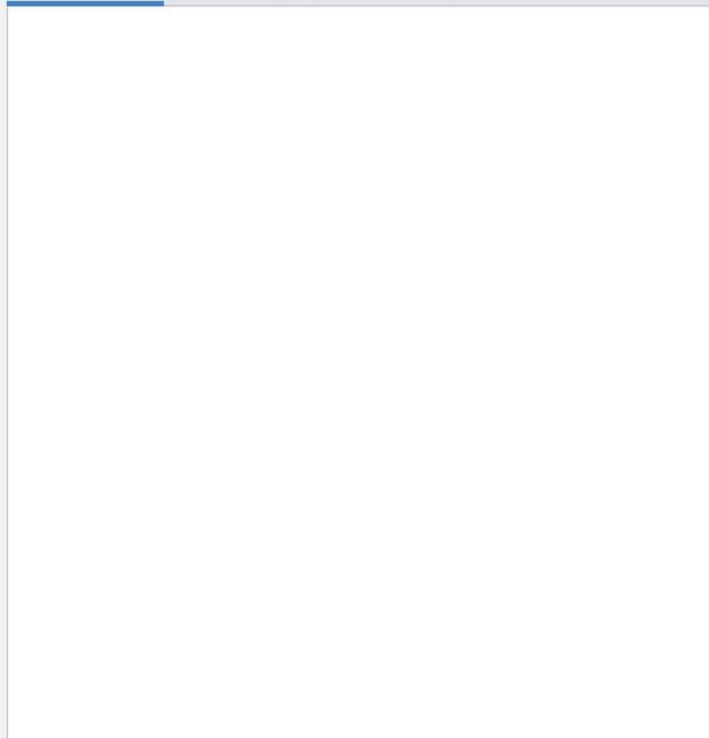




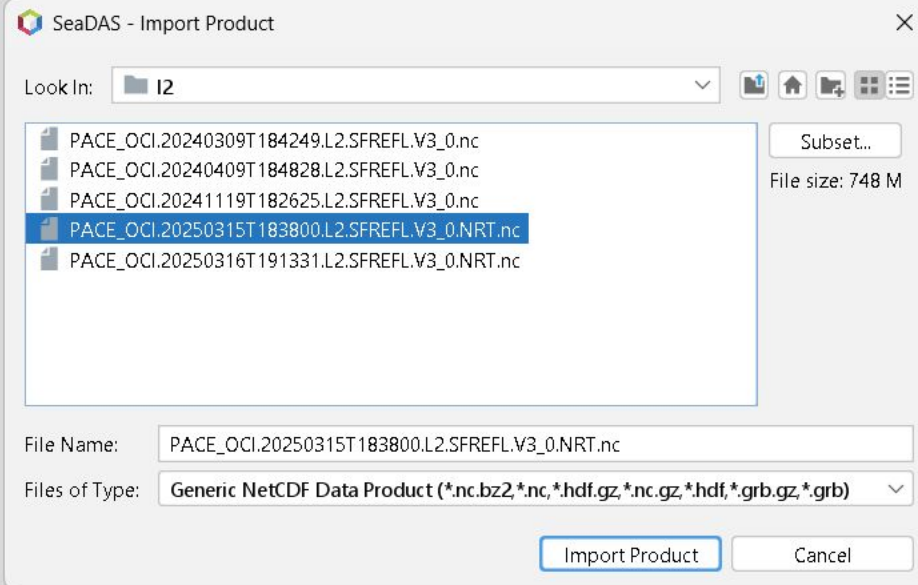
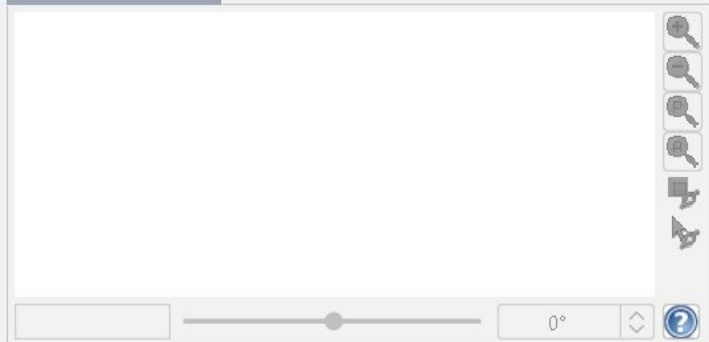


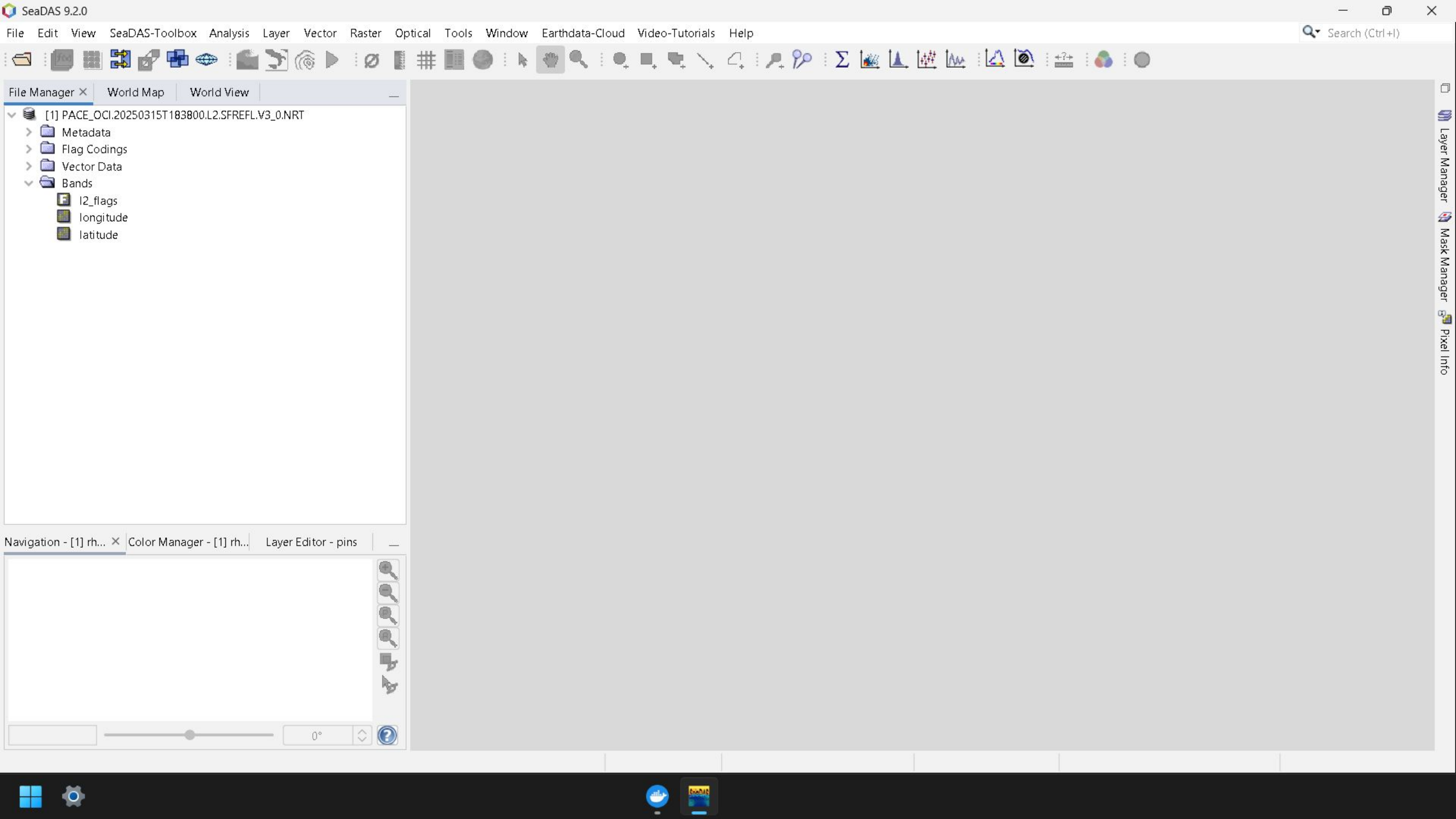


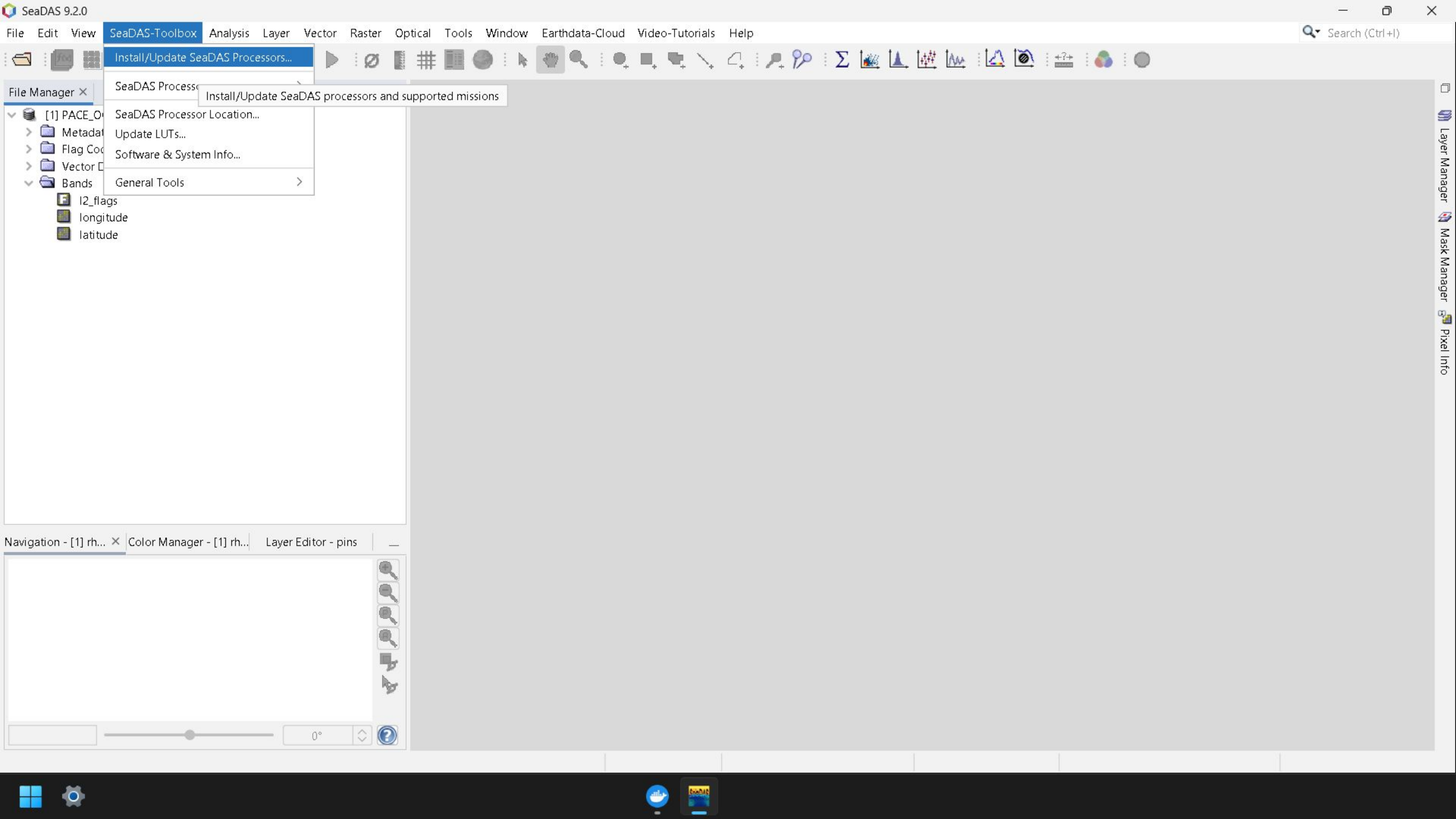
File Manager × World Map World View

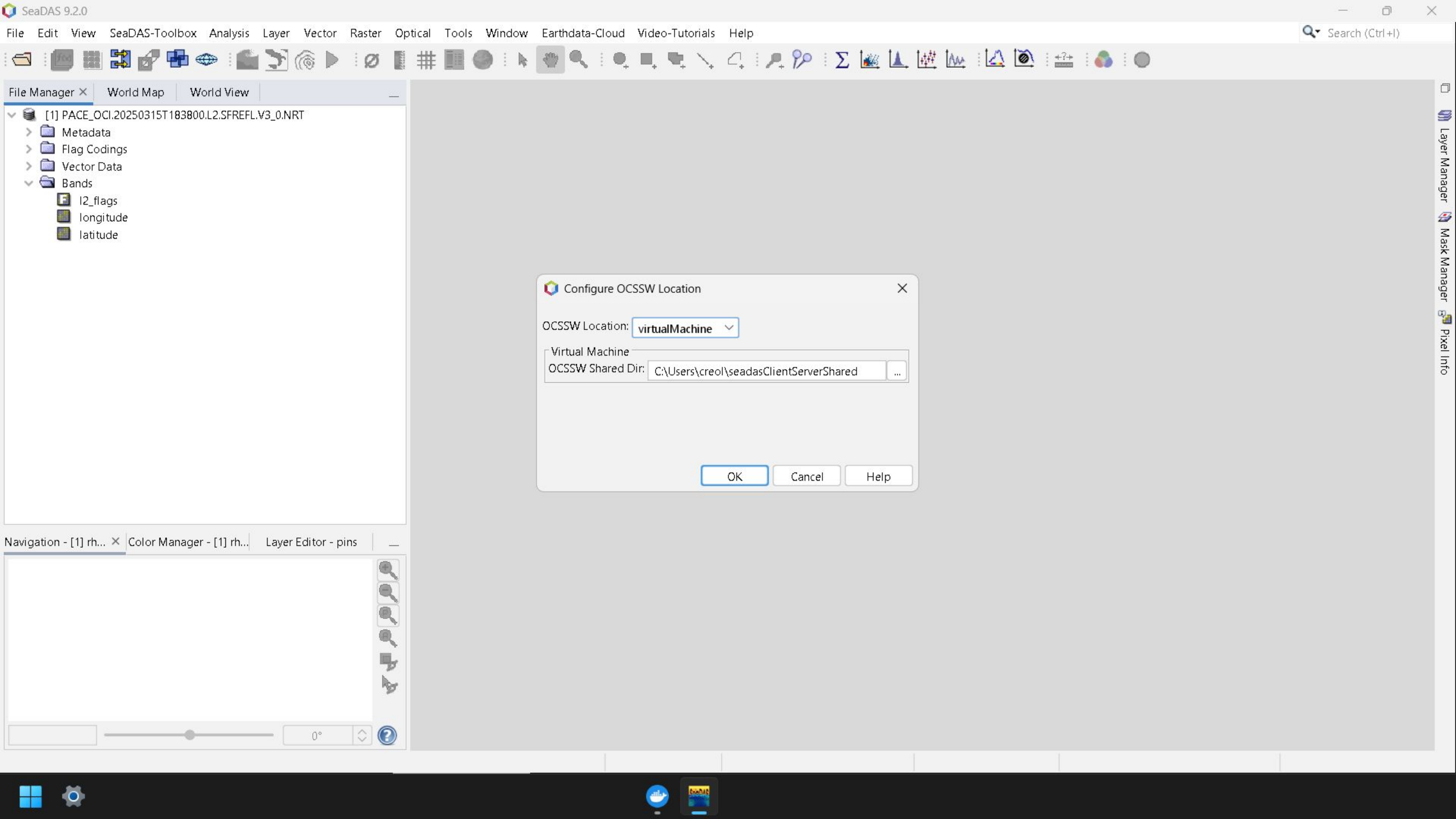


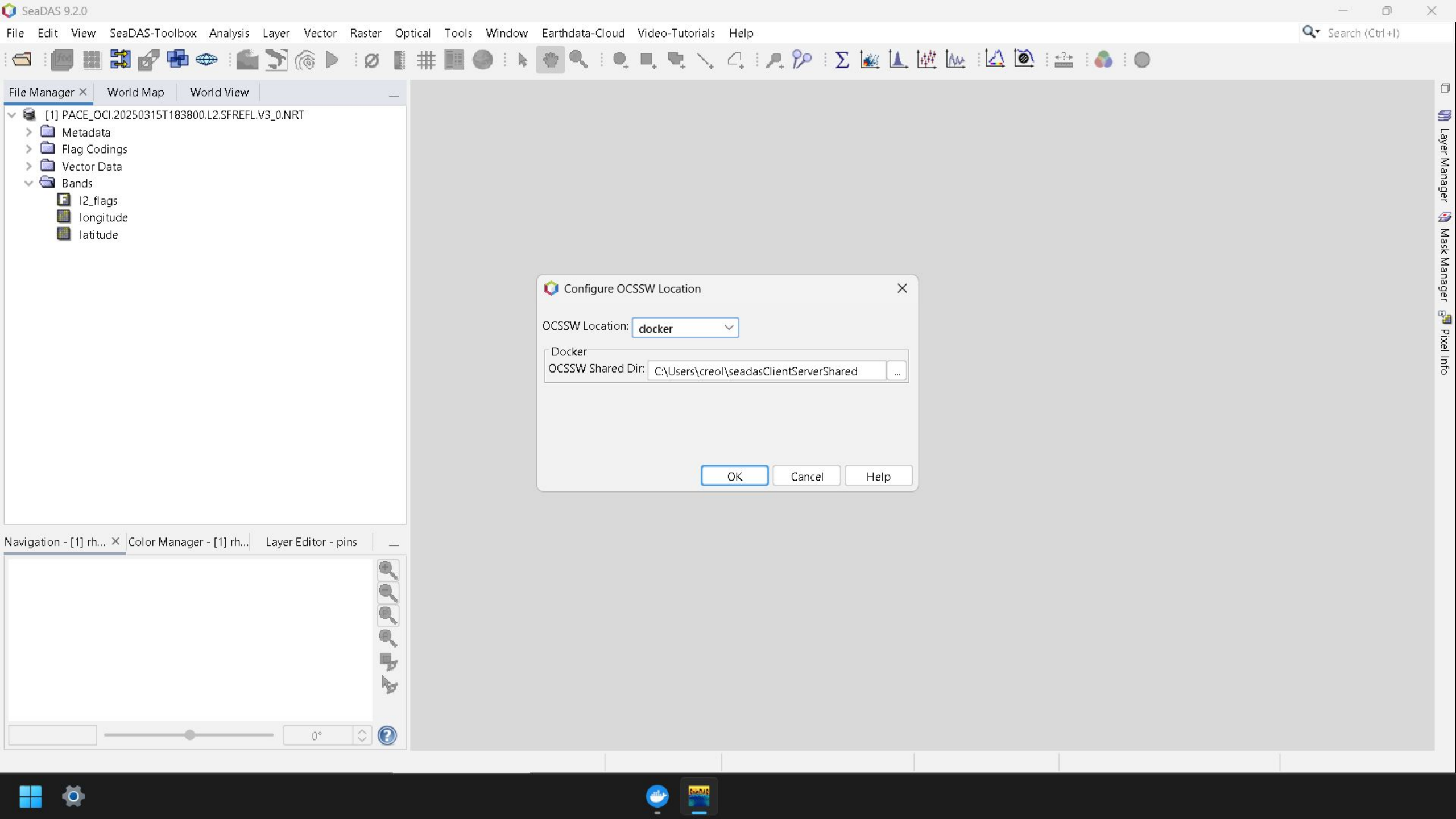
Navigation - [1] rh... × Color Manager - [1] rh... Layer Editor - pins

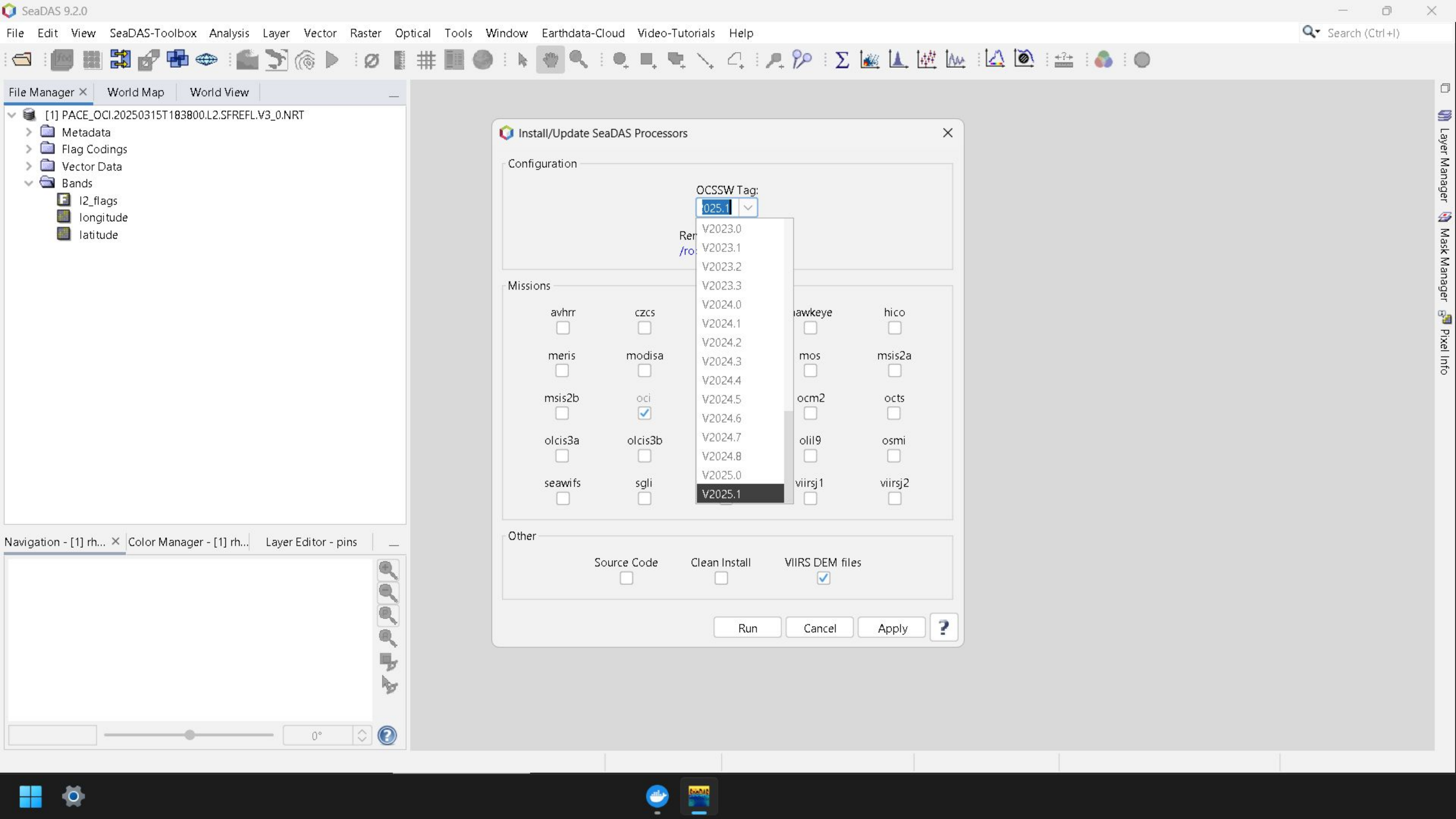


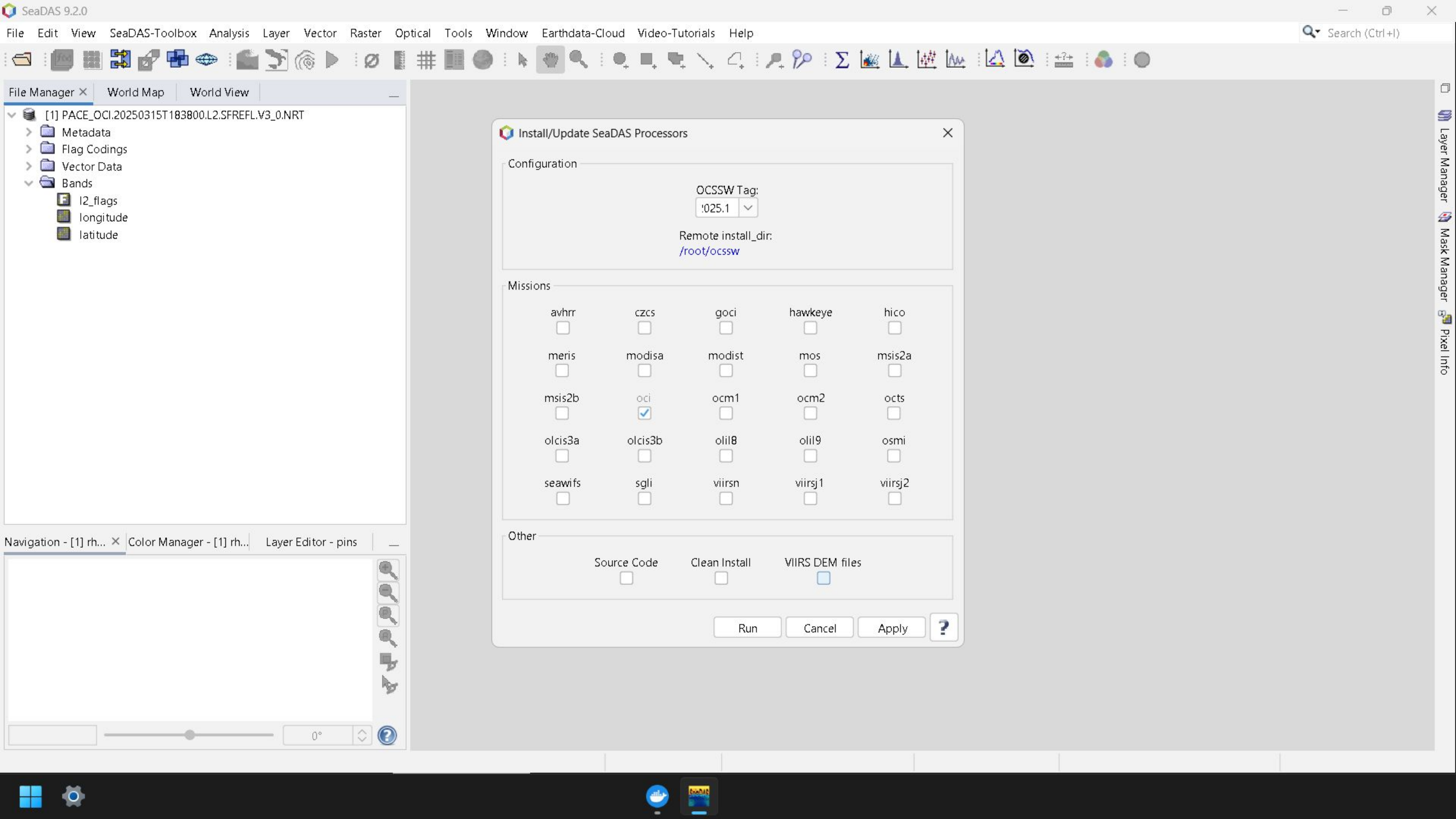


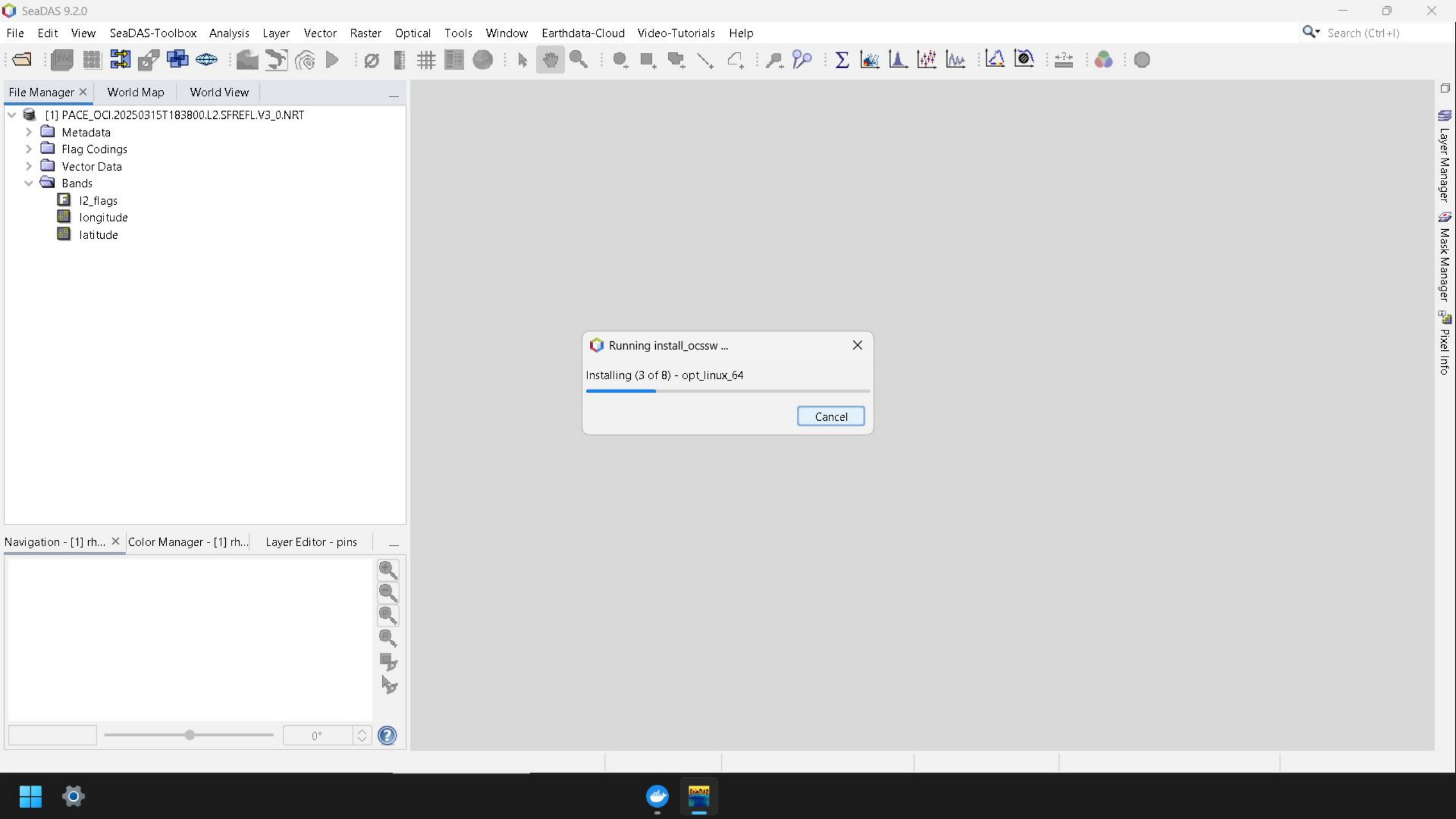


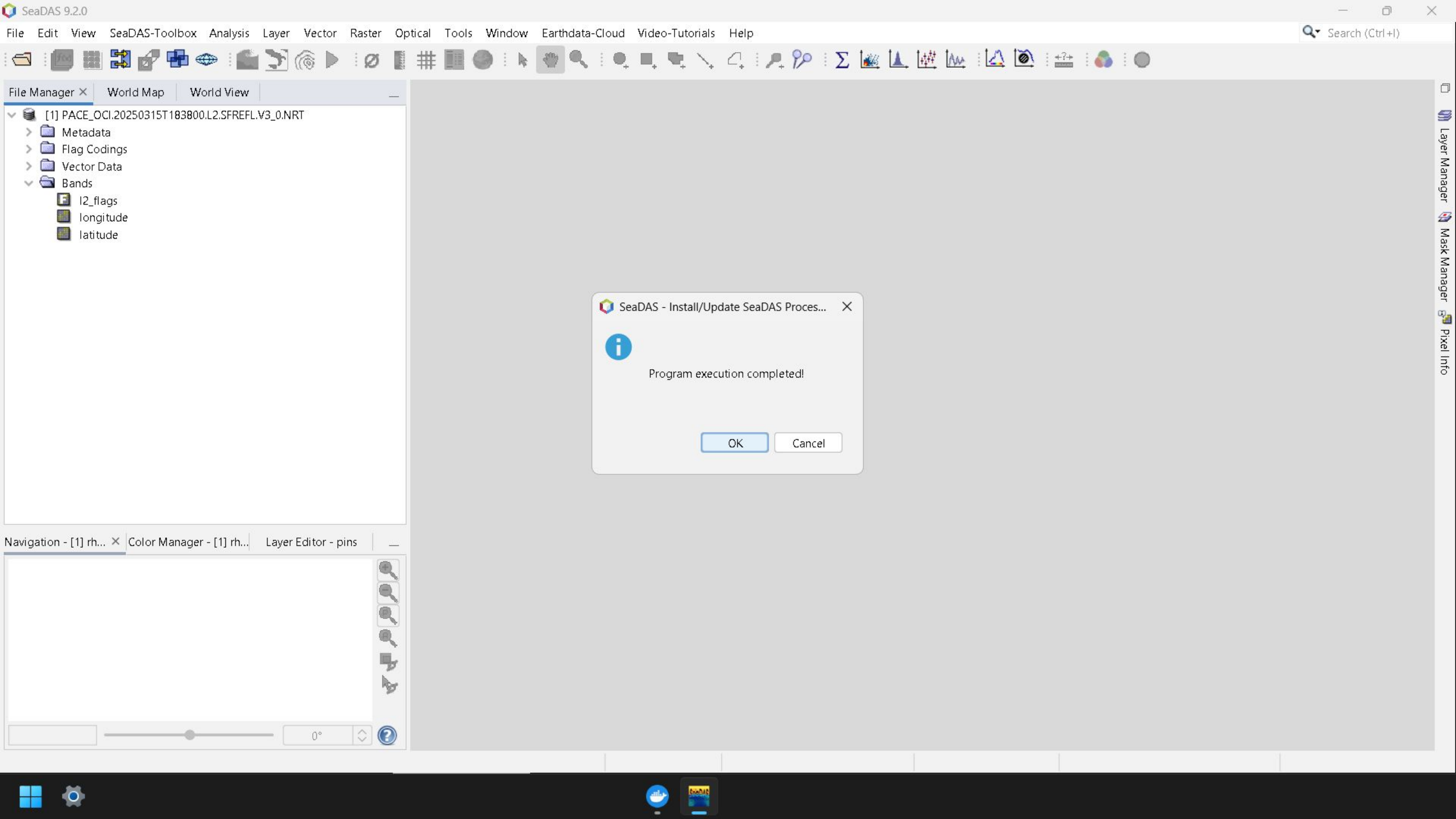


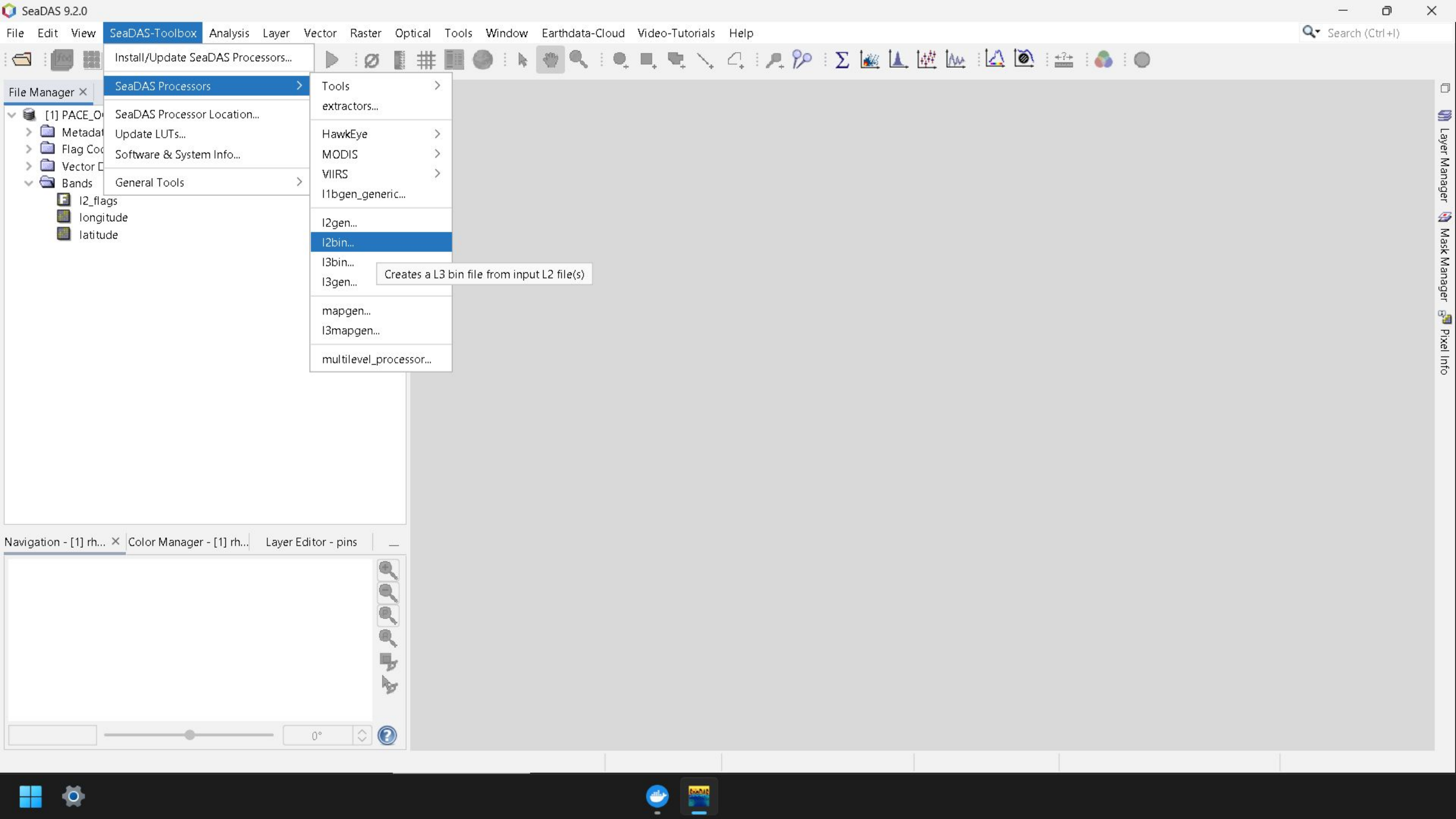


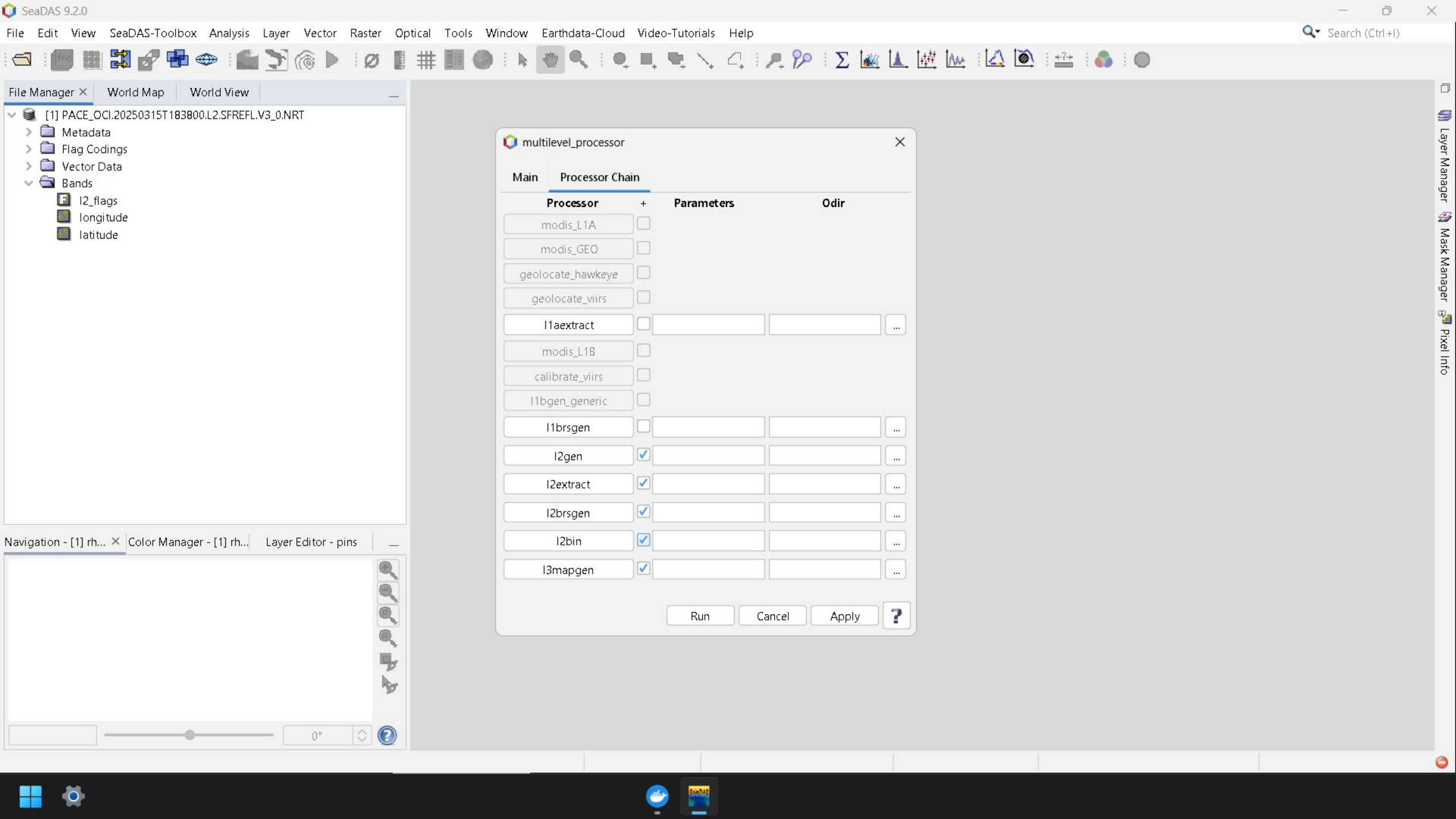


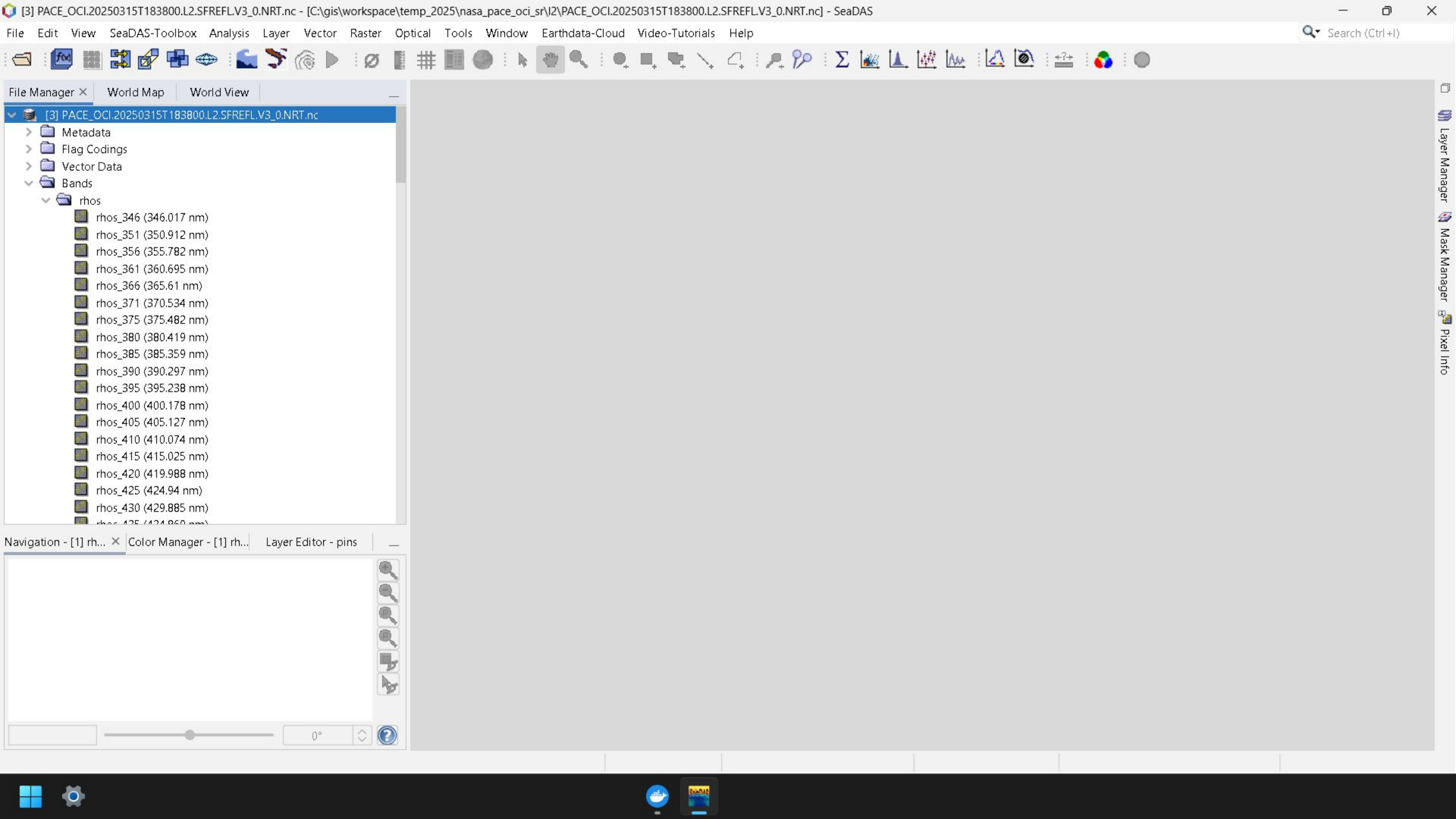


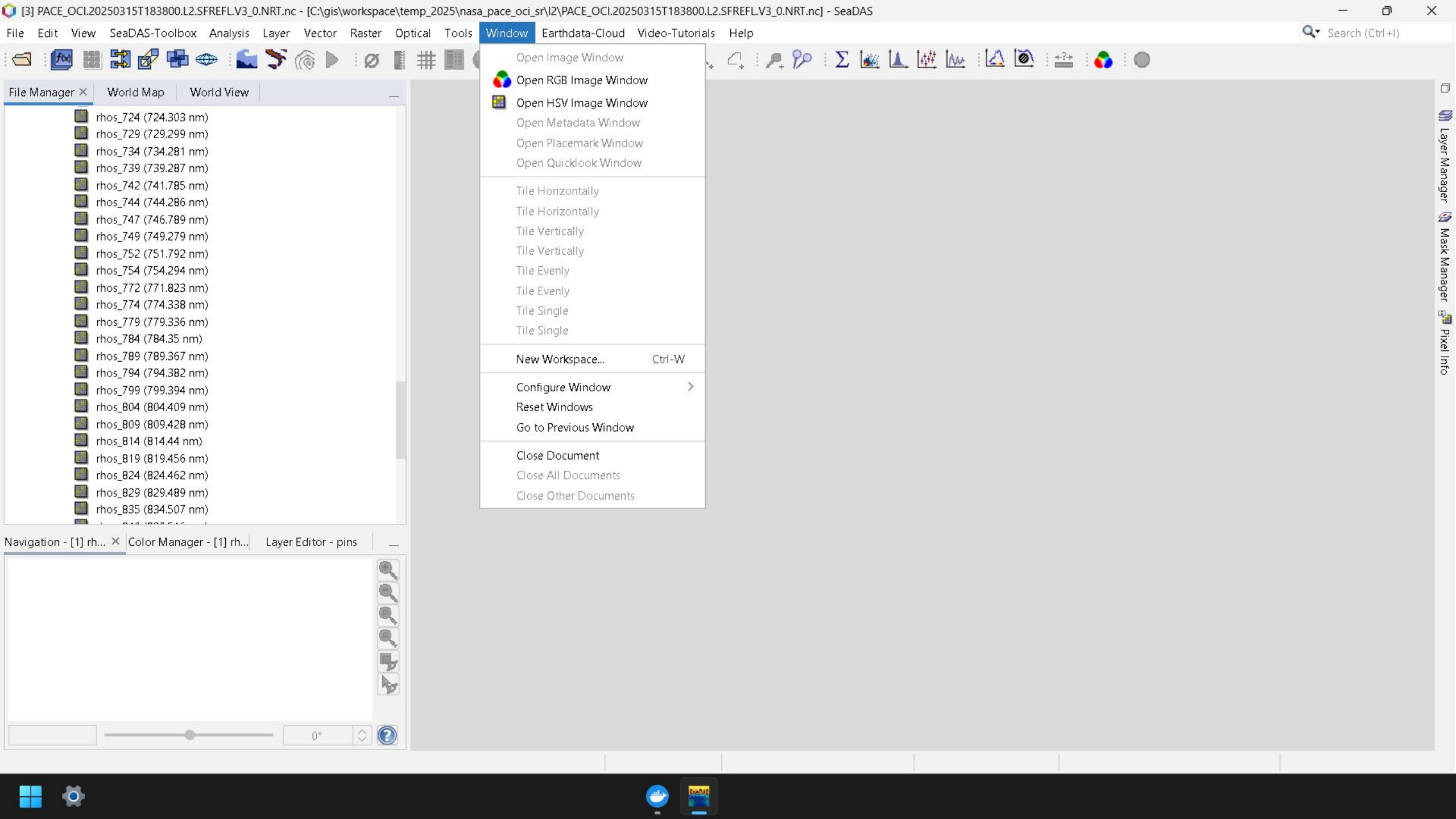


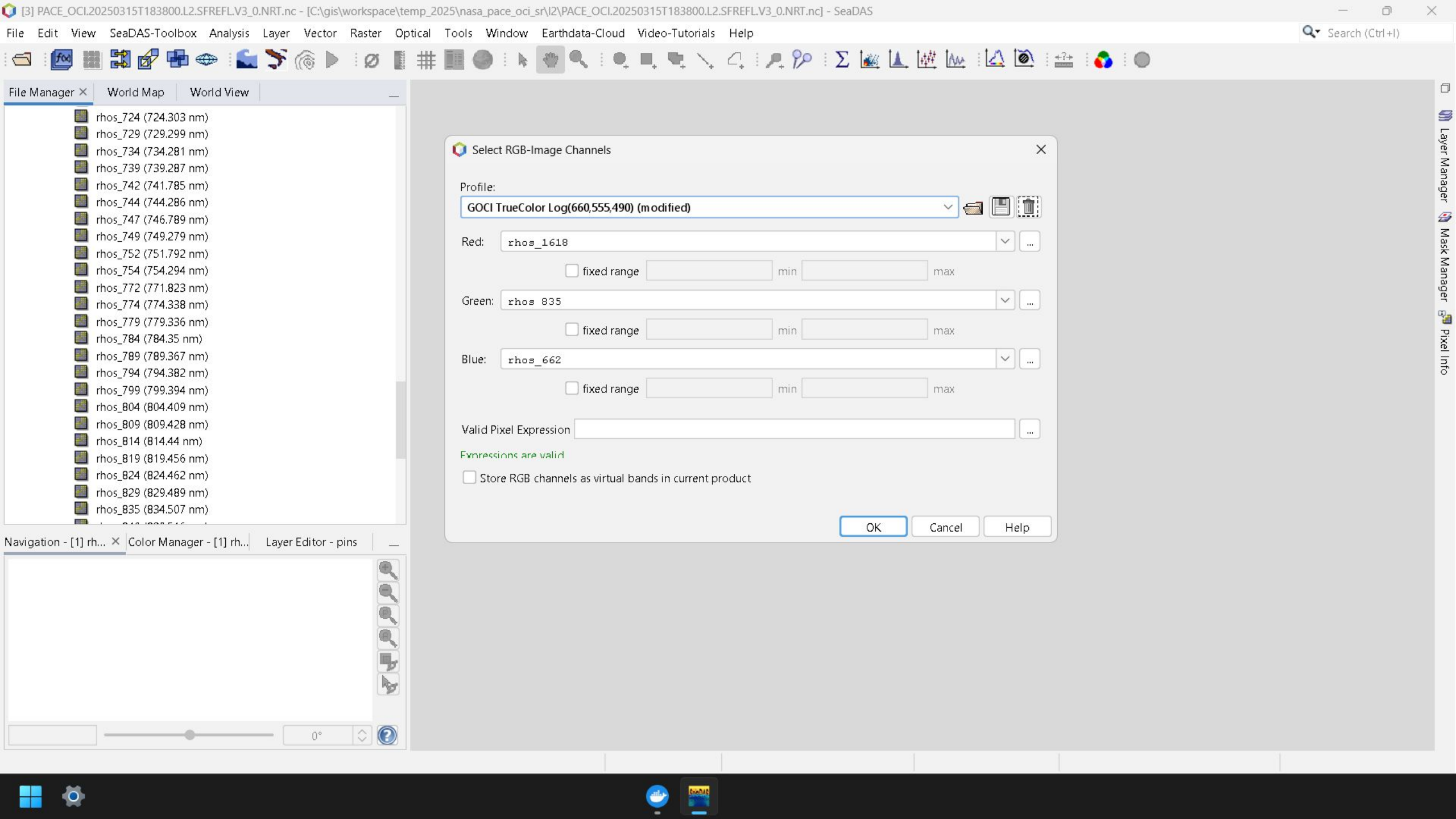


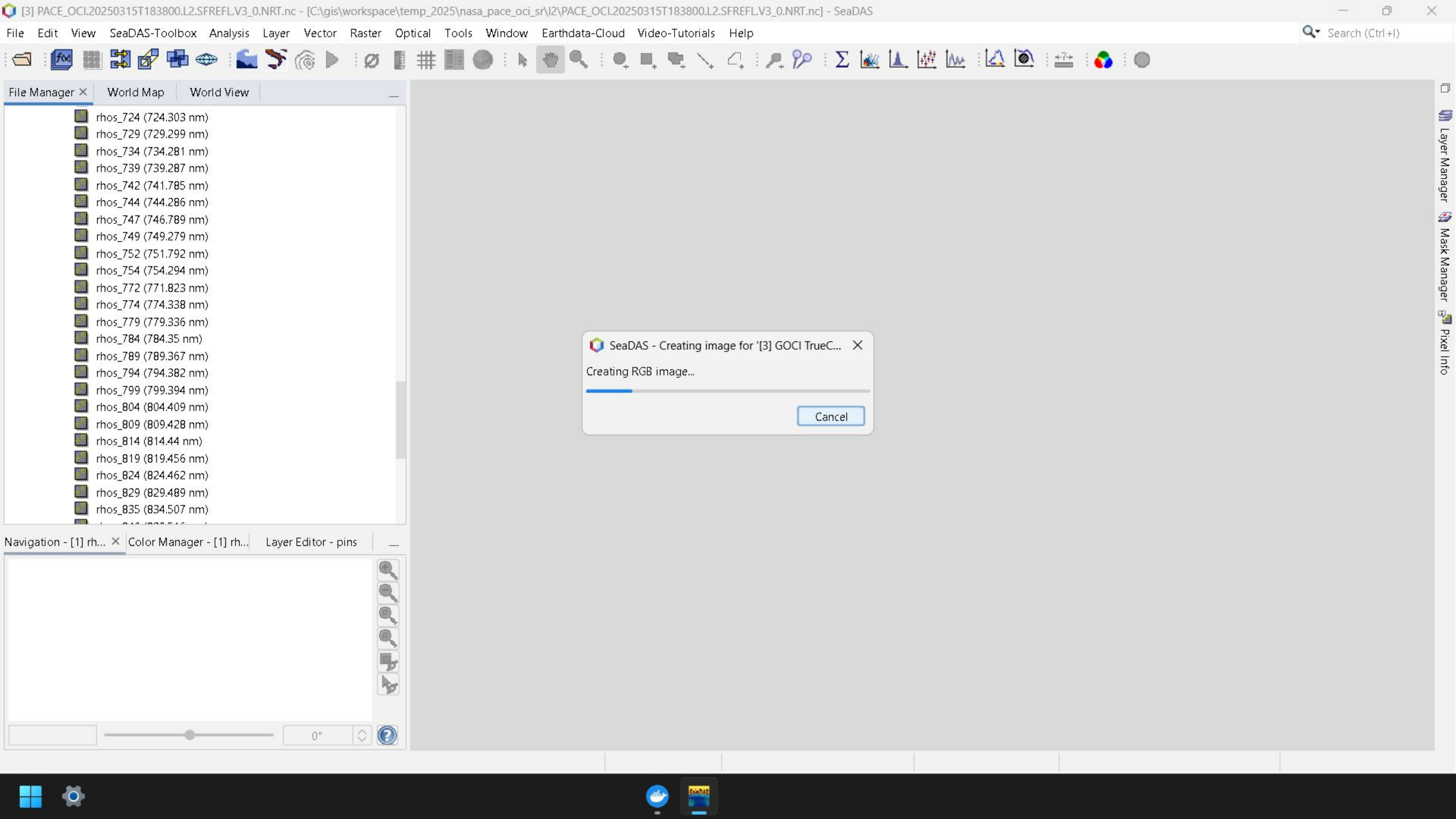


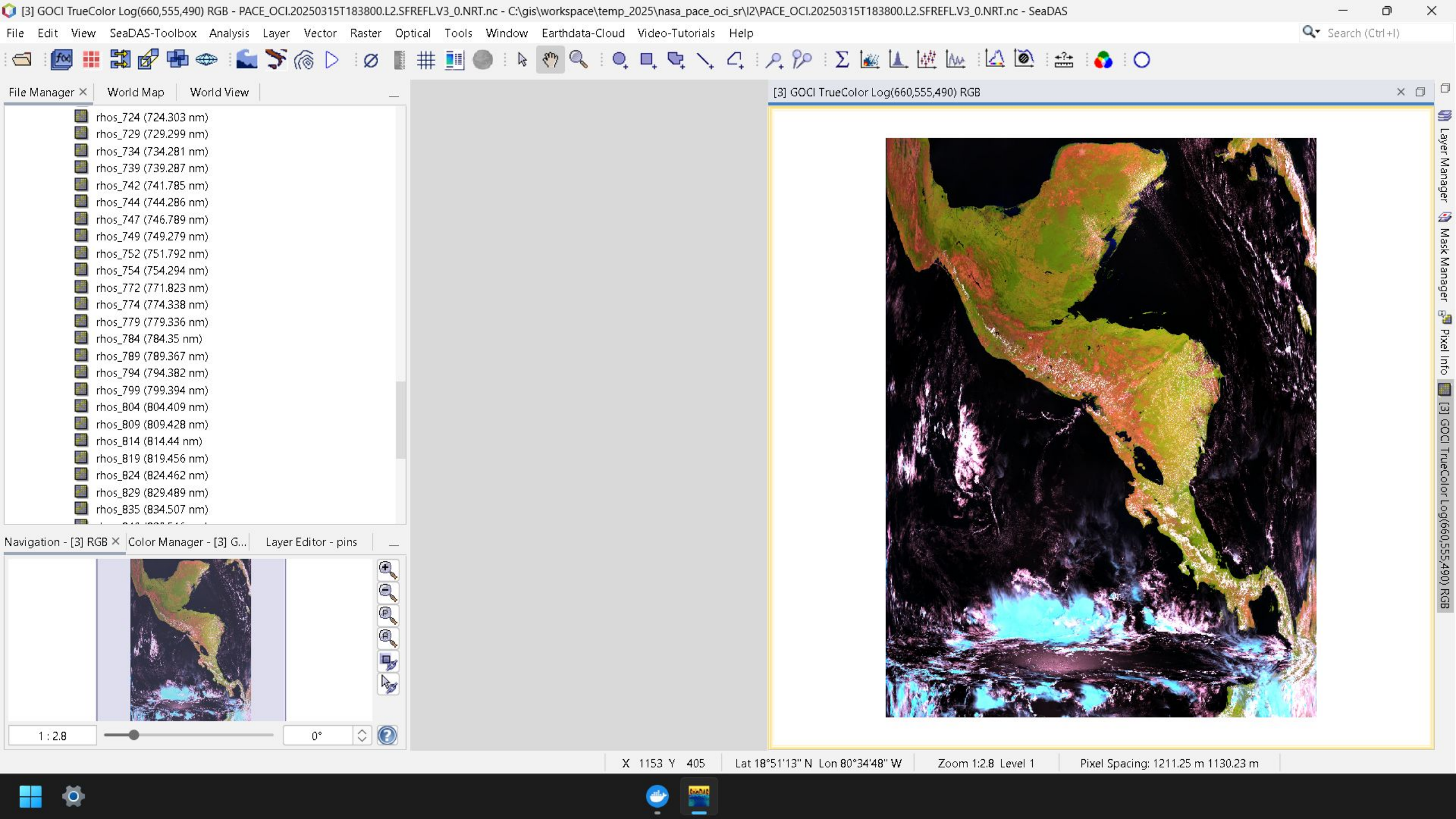


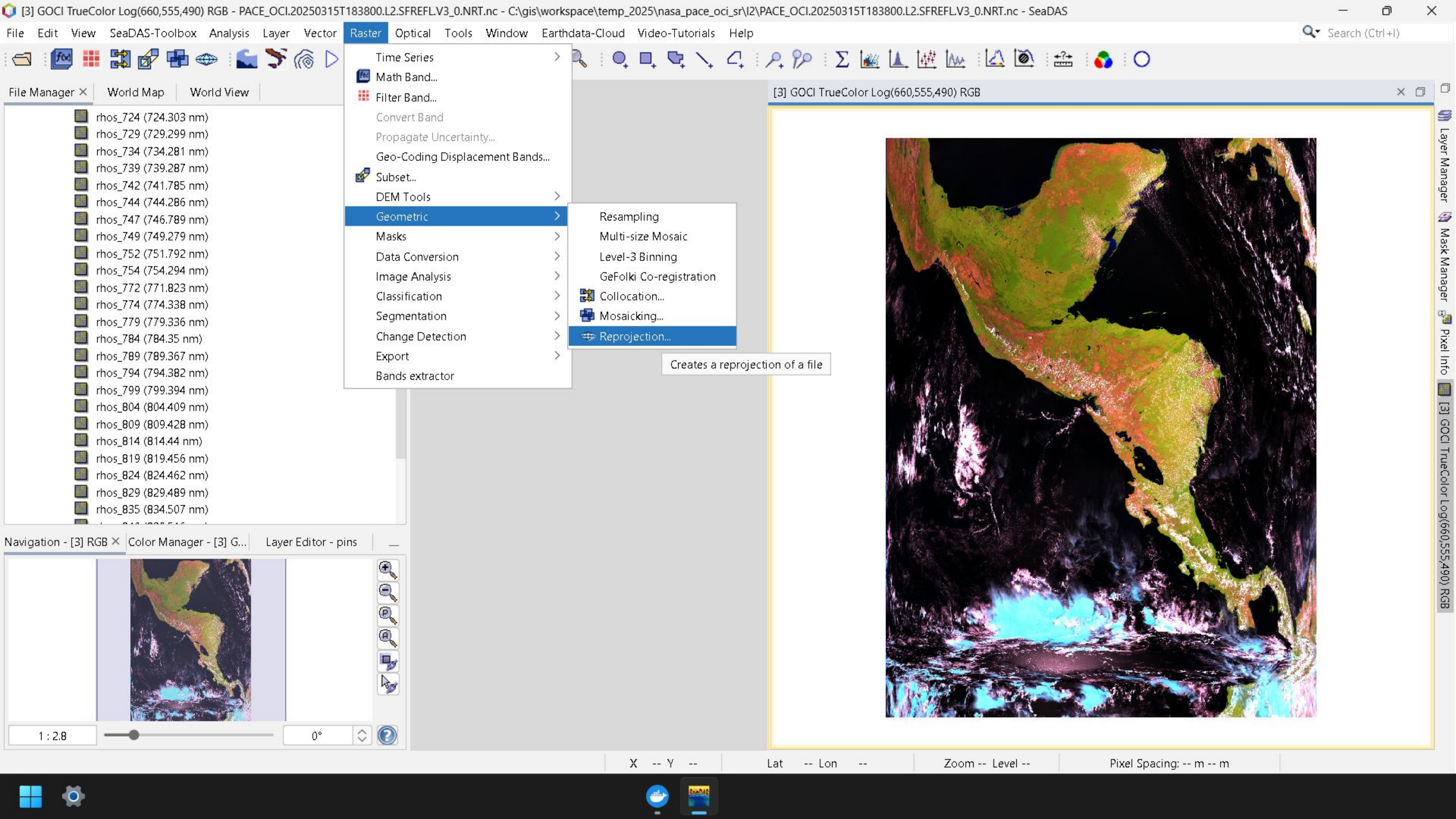












File

Edit

View

SeaDAS-Toolbox

Analysis

Layer

Vector

Raster

Optical

Tools

Window

Earthdata-Cloud

Video-Tutorials

Help

Search (Ctrl+I)

File Manager

World Map

World View

rhos_724 (724.303 nm)

rhos_729 (729.299 nm)

rhos_734 (734.281 nm)

rhos_739 (739.287 nm)

rhos_742 (741.785 nm)

rhos_744 (744.286 nm)

rhos_747 (746.789 nm)

rhos_749 (749.279 nm)

rhos_752 (751.792 nm)

rhos_754 (754.294 nm)

rhos_772 (771.823 nm)

rhos_774 (774.338 nm)

rhos_779 (779.336 nm)

rhos_784 (784.35 nm)

rhos_789 (789.367 nm)

rhos_794 (794.382 nm)

rhos_799 (799.394 nm)

rhos_804 (804.409 nm)

rhos_809 (809.428 nm)

rhos_814 (814.44 nm)

rhos_819 (819.456 nm)

rhos_824 (824.462 nm)

rhos_829 (829.489 nm)

rhos_835 (834.507 nm)

Navigation - [3] RGB

Color Manager - [3] G...

Layer Editor - pins

1 : 2.8

0°

Reprojection

File

Help

I/O Parameters

Reprojection Parameters

Source Product

Name:

[3] PACE_OCI.20250315T183800.L2.SFREFL.V3_0.NRT.nc

Target Product

Name:

PACE_OCI.20250315T183800.L2.SFREFL.V3_0.NRT.nc_reprojected

☒ Save as:

BEAM-DIMAP

Directory:

C:\gis\workspace\temp_2025\nasa_pace_oci_sr\l3b_reproj_gcs

☒ Open in SeaDAS

Run

Close

55,490) RGB

Layer Manager

Mask Manager

Pixel Info

[3] GOCI TrueColor Log(660,555,490) RGB

X -- Y --

Lat -- Lon --

Zoom -- Level --

Pixel Spacing: -- m -- m

